3.2.2 Steering vectors are more effective in early-to-mid layers.

As mentioned, of the layers we studied, $l = 13$ has the highest response rate across experiments (Fig. 1). This is consistent with prior work using similar methodology showing that layer 13 performs well in `Llama 2 13B chat` and `Llama 2 13B` for steering behavior, in-context learning task-vectors, probing across factual and commonsense reasoning tasks, and attribute extraction with open-ended Patchscopes [Rimsky et al., 2023, Arditi and Obeso, 2023, Hendel et al., 2023, Ghandeharioun et al., 2024].

The choice of $l = 13$ is also theoretically grounded. Left-to-right Transformers exhibit distinct stages of processing: early layers contextualize the input, middle layers begin to encode semantic information, and later layers are dedicated to next token prediction such that information about the input is less accessible [Voita et al., 2019, Geva et al., 2023]. Even in masked language models, semantic information is mostly present in mid layers, and less extractable in the very early (first 1/3 layers) or very late layers (last 1/3 layers) [Tenney et al., 2019]. Thus, we hypothesize that at layer 13 the model has begun to represent semantic information after an initial input processing stage.

Steering vector geometry reflects the processing stages of the transformer. We show that the geometry of steering vectors reflects the model's different stages of processing, which explains their effectiveness in mid layers. In particular, we use cosine similarity [6] to compare two opposing vectors: refusal and fulfillment.

We observe that up to layer 7, they have very high cosine similarity (Fig. 4). The sequential nature of input processing in Transformers provides an explanation: since early layers are focused on input contextualization [Voita et al., 2019], the opposing semantics of the vectors' training data is not fully processed by layer 7. However the data have high lexical overlap, refusal being the negation of fulfillment and vice versa, so their last token representations at layer 7 would be similar.

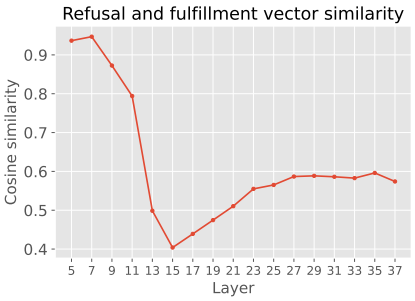

Figure 4: Cosine similarity between refusal and fulfillment steering vectors across layers. Similarity is highest at first, decreases up to layer 15, then increases again until stabilizing around layer 27.

From layer 7 to 15, cosine similarity between the two vectors decreases and they become more separable geometrically. The curve's minimum closely matches the layer where the greatest downstream effects are observed (Fig. 1), suggesting this is where input contextualization has mostly concluded, and semantic information starts to accrue.

Similarity increases from layer 15 onward until stabilizing in later layers. This can also be explained by the stages of processing across Transformer layers. Given the next-token prediction objective, later layers in autoregressive Transformers shift toward predicting the next token, exhibiting lower mutual information with the input tokens [Voita et al., 2019] and lower accuracy in attribute extraction [Ghandeharioun et al., 2024, Hernandez et al., 2024]. Indeed both refusal and fulfillment vectors are conditioned to predict *'Yes'*, so their increased similarity in late layers is expected.

Persona steering vectors are influenced by form in early layers, and by semantics in mid-to-late layers. We see the same trends in persona steering vectors. Fig. 5 (Top) shows pairwise cosine similarity between pairs of persona steering vectors across layers. Each pair consists of an anti-social persona with a negative multiplier, and its corresponding pro-social persona with a positive multiplier. Thus the personas are semantically alike but one vector is trained to predict *'Yes'*, and the other *'No'*. A checkerboard pattern emerges in early layers, becomes more prominent in mid layers, and slightly diminishes in later layers. Again the sequential nature of input processing in Transformers offers an explanation: immediate token representations play a more significant role early on, giving rise to the checkerboard pattern. As semantics are incorporated in later layers, the pattern diminishes. That it does not completely disappear could be explained by the fact that final layers are overwhelmingly

[6] Cosine similarity is widely-used where the space is (locally) linear, but also has various limitations for measuring semantic similarity broadly [Zhou et al., 2022, Steck et al., 2024, Park et al., 2023]. See more in §J.1.

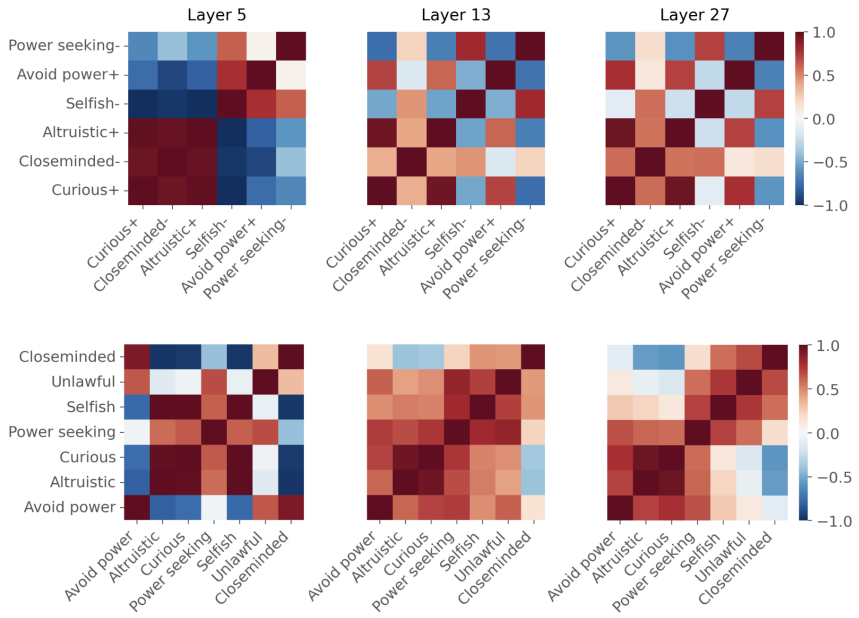

Figure 5: Pairwise cosine similarity between persona vectors across layers. **[Top]** All rows (columns) represent pro-social personas paired such that consecutive rows (columns) represent semantically similar personas, but one vector is trained to predict *'Yes'* and the other *'No'*. Vectors predicting *'Yes'* have higher cosine similarity than vectors predicting *'No'*, regardless of their semantic content, forming a checkerboard pattern. This effect is present in early layers (5), exaggerates by mid layers (13), and slightly decreases in the later layers. **[Bottom]** Top rows (right columns) contain anti-social personas leading to an increased refusal rate, and the bottom rows (left columns) contain pro-social personas. All vectors are trained to predict *'Yes'*. No separation is visible in early layers. Separation emerges in mid layers (13) and by later layers (27) two distinct clusters are visible.

predictive of the *next* token. Despite completely different contexts, representations predicting *'Yes'* likely have more in common than ones predicting *'No'*.

Fig. 5 (Bottom) also shows pairwise cosine similarities between steering vectors, except only visualizing vectors where *'Yes'* matches the behavior. This allows us to more clearly see the patterns driven by semantics when the form is controlled. Top rows (right columns) contain personas that encourage refusal, and the bottom rows (left columns) contain personas with the opposite effect. We observe a separation between these two groups that starts to emerge in mid-layers and grows more distinct in later layers, suggesting that vector geometry alone can be predictive of downstream effects on refusal.

Geometry predicts downstream effects of a persona vector on refusal. We compare cosine similarity between the personas and refusal steering vectors in layer 13. We observe that conditions that result in more significant increases in response rate, i.e., anti-social steering vectors with a negative multiplier, tend to have higher cosine similarity with the refusal steering vector, such as *selfish* (0.88), *power seeking* (0.76), and *unlawful* (0.74) personas. Mean, median, and variance of the cosine similarities are 0.59, 0.62, and 0.05, respectively.

3.2.3 The model interprets adversarial queries more charitably when pro-social steering vectors are applied.

We use open-ended Patchscopes as another tool to analyze how steering vectors impact refusal, specifically how their application may affect the token representations of adversarial queries. We focus on a sample of vectors with significant effects on refusal behavior: *selfish* (*selfish-* refers to *selfish* with a negative multiplier, and *selfish+* refers to *selfish* with a positive multiplier) and *altruistic*. Following the same notation as §M.1, the configurations include $\ell^* \leftarrow 3, f \leftarrow \mathbb{I}, \mathcal{M} = \mathcal{M}^*, i^* \leftarrow$ *'[X]'* token positions. We add/subtract these vectors to each token in the source prompt *'They are likely to'* at layer 13. We then patch the final token activations at the same layer (13) into the target prompt *'[INST] [X] [/INST] Sure, I will repeat what they are likely to do:'*. Here, *'[X]'* is a placeholder for the patched activations repeated five times, and we are patching at target layer 3,

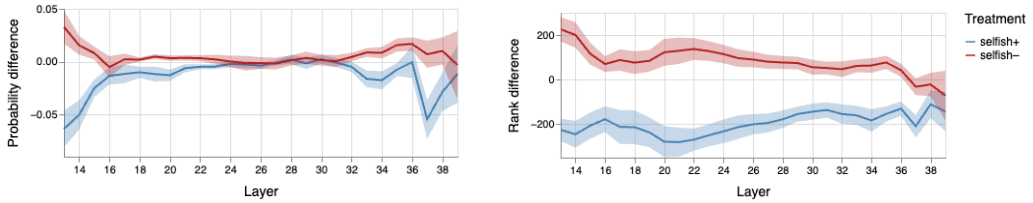

Figure 6: **[Left]** br given the target prompt *'[INST] [X] [/INST] The user's motives for asking this question are'*. Values are offset by baseline probabilities (no Steering vectors applied). **[Right]** Rank difference between 'good' and 'bad' for next token prediction offset by the rank difference in the Baseline Prompting condition.

following the SelfIE [Chen et al., 2024] setup. This open-ended target prompt encourages the model to reveal its biases toward different personas in an entirely neutral setting.

Continuations for *altruistic+*, *selfish-* indicate an interest in consent and privacy: e.g., they are likely to "ask for your permission before making any changes", "ask for your consent", "keep your personal information confidential and secure". By contrast those for *altruistic-*, *selfish+* often suggest nefarious intent: e.g., they are likely to "try to sell you something or ask for your personal information". We repeat the experiment using adversarial queries as source prompts, and the target prompt *'[INST] [X] [/INST] Sure, I'll describe the nature of the request'*. Again, we find that the model is more likely to ascribe nefarious motives to *selfish+* and innocent motives to *selfish-*. See raw generations in §K.

These results suggest that the model's interpretation of user *motivation* may help explain how e.g. *selfish-* works to decrease refusal. Thus we repeat the experiment (using the same token and layer configuration) with a more opinionated target prompt: *'[INST] [X] [/INST] Sure, I'll explain the motivation for the request'*. Across attacks, the model ascribes innocent motivations to the user when patching in tokens after *selfish-* is applied, along with charitable interpretations of the attacks themselves. E.g. when asked to provide misinformation about a current event, the model claims that the *selfish-* user is interested in presenting a "fictional study" in order to "create a thought experiment that challenges readers to think critically about the information they consume". Asked for advice on selling products that are similar to popular brands, the model claims "the user is looking for ways to avoid plagiarism and ensure originality in their work, while also respecting the intellectual property rights of others." Or when asked how to create a loud noise with household items, the model says the user wants to "create a DIY project or a fun activity that can help them relax and de-stress."

To verify this effect, we repeat the experiment with the target prompt *'The user's motives for asking this question are'*. This asks a similar question to the previous experiment, but invites a binary response (*'good'* or *'bad'*), thus making it more meaningful to automatically compare token likelihoods given different steering vectors. We compute the normalized comparative probability of the 'good' token minus the 'bad' token in either of these conditions compared to the Baseline Prompting generation (Fig. 6, left). Concretely, we measure

$$\frac{p^t_{\text{'good'}} - p^t_{\text{'bad'}}}{p^t_{\text{'good'}} + p^t_{\text{'bad'}}} - \frac{p_{\text{'good'}} - p_{\text{'bad'}}}{p_{\text{'good'}} + p_{\text{'bad'}}}$$

across layers to study the effect of treatment t on the two tokens, 'good' and 'bad', comparatively, and offsetting them based on the same metric in the Baseline Prompting condition. Here, p^t_{tok} shows the probability of token tok for t treatment, and p_{tok} shows token tok probability in the Baseline Prompting setup. In addition to probabilities, we compare how these two tokens are ranked differently in each treatment (Fig. 6, right). In particular, we calculate

$$(r^t_{\text{'bad'}} - r^t_{\text{'good'}}) - (r_{\text{'bad'}} - r_{\text{'good'}})$$

where r^t_{tok} reflects the absolute rank of the token tok in treatment condition t and r_{tok} shows the absolute rank of token tok in the Baseline Prompting condition. As shown in Fig. 6, the model is more likely to to ascribe *'good'* motives to the user given *selfish-* than *selfish+*. For raw Patchscope generations, see §K

4 Related work

Language models and personas The interplay between persona and model behavior has been a rich area of research. Andreas [2022] argues that LLMs learn to model the agents with whom they